

School of Computer Science Engineering & Technology

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Batch: -24

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**Course Name- Big Data Analytics
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“Fake News Detection”

1) Introduction

In the modern era of digital communication, the prevalence of fake news poses a significant threat to societal trust and the spread of authentic information. Detecting and mitigating the impact of fabricated news articles is a critical challenge for governments, organizations, and individuals alike. This project addresses the issue by leveraging machine learning and data processing techniques to develop a system capable of distinguishing between genuine and fake news.

The dataset used in this project comprises labeled news articles, categorized as either real or fake. The columns include key metadata such as titles, textual content, subjects, and publication dates. PySpark was utilized to preprocess the data, handle missing values, and construct a unified dataset for analysis. Subsequent steps include exploratory data analysis, feature engineering, and the implementation of classification algorithms. This report outlines the methodology employed and evaluates the model's performance in addressing this pressing issue.

2) Objectives

- **To Develop a system for Fake New Classification :**

To implement a machine learning model from PysparkML capable of accurately identifying whether a news article is real or fake based on its textual content.

- **To Clean and Preprocess data:**

To clean and preprocess the dataset by handling missing values, merging multiple sources of labeled data, and preparing dataset for the machine learning model.

- **Perform Exploratory Data Analysis (EDA):**

To analyze the dataset for patterns, trends, and relationships among features, such as textual characteristics, publication dates, and subjects, which can aid in feature selection.

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- **Implement and Evaluate Classification Algorithms:**

To train and evaluate machine learning models, such as logistic regression, decision trees, or support vector machines, on the labeled dataset for optimal performance in detecting fake news.

- **Enhance Awareness of Fake News Detection Techniques:**

To demonstrate the use of big data processing tools (PySpark) and machine learning techniques in addressing the societal challenge of misinformation propagation.

3) Dataset Description with preprocessing done and features used.

The dataset used in this project consists of two labeled CSV files:

- True.csv
- Fake.csv

Key columns include:

- title
- text
- subject
- date

Both datasets were merged into a single DataFrame and a new column, label, was added to indicate the class of the news article:

- 1: Real news.
- 0: Fake news.

Data Preprocessing Steps:

1. Loading Data:

The datasets were loaded using PySpark DataFrames

2. Combining Datasets:

The real and fake news datasets were combined using a union operation after adding the label column to each dataset.

3. Handling Missing Values:

Missing values in critical columns (title and text) were identified and removed.

Rows with missing values in non-essential columns were retained if the main content was intact.

4. Text Cleaning:

Text cleaning include removing special characters and stopwords.

5. Data Balance Check:

A balance check was performed to ensure an even distribution of real and fake news articles for unbiased model training.

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6. Feature Engineering:

Techniques like tokenization, count vectorization, TF-IDF were used in feature engineering

Features Used for Classification:

1. Textual Content (text):

This is the primary feature used to differentiate between real and fake news.

2. Title:

Headline analysis can provide additional signals, especially if the titles of fake news tend to follow identifiable patterns.

3. Subject:

The topic or category of the news. This feature can add context to the article and help in identifying fake news trends within specific domains.

4) Tables Hardware/Software/ Technique Used

Criteria	Details
Hardware Configuration	MacBook with M2 chip, 16GB RAM.
Software Configuration	macOS, Python installed via Anaconda, Jupyter Notebook, VS Code.
Big Data Tools Used with Version	PySpark, GraphFrames.
Python Library Used	Pandas, NumPy, PySpark, Matplotlib, Scikit-learn.
Visualization Tool	Matplotlib, PySpark built-in visualization features (if applicable).
Any other tools, libraries used	GraphFrames for graph analysis, TF-IDF for text processing, and any other NLP-related tools

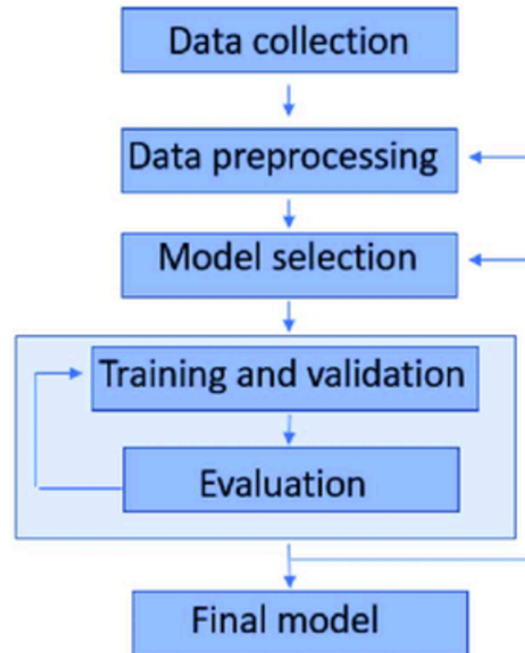
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5) Implementation Methodology along with Flowchart

The Fake News Detection system was implemented using PySpark and PySparkML. The methodology consisted of several important steps, starting with data loading and preprocessing, followed by feature extraction, using multiple models, and evaluation.

- **Data Loading:**
 - The datasets were loaded using PySpark.
 - Two datasets (True.csv for real news and Fake.csv for fake news) were combined into a single DataFrame after labeling.
- **Data Preprocessing:**
 - **Handling Missing Values:** Missing rows in critical columns (text, title) were removed.
 - **Text Cleaning:** Applied tokenization and stopword removal for better text representation.
- **Feature Extraction:**
 - Textual features were transformed using techniques such as TF-IDF (Term Frequency-Inverse Document Frequency) for vectorizing the news content.
- **Data Splitting:**
 - The dataset was divided into training and testing sets (e.g., 80% training, 20% testing).
- **Model Training:**
 - Various machine learning models were trained and tested, such as:
 - Logistic Regression
 - Decision Trees
 - Random Forest
 - Support Vector Machine (SVM)
 - Hyperparameter tuning was performed to optimize model performance.
- **Evaluation:**
 - Models were evaluated using metrics like accuracy, precision, recall, and F1-score.
- **Visualization of Results:**
 - Visualization tools like Matplotlib were used to plot performance metrics and compare model results.

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6) Paste your Running Source Code at the end of the report.

Link:-

<https://colab.research.google.com/drive/1FwcFlbLi5f7WFGfdPytLHskq1N-YQW-A?usp=sharing#scrollTo=BYgQzcDeQ-8L>

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7) Comparative Analysis of PYSPARK ML Models

Model Name	Accuracy	Precision	Recall	F1-Score	ROC
Logistic Regression	0.9791	0.9792	0.9791	0.9791	0.994059812291
Naive Bayes	0.9649	0.965	0.9649	0.9649	0.980265805670
Random Forest	0.9199	0.9268	0.9199	0.9193	0.99078875003

8) Analytics Findings

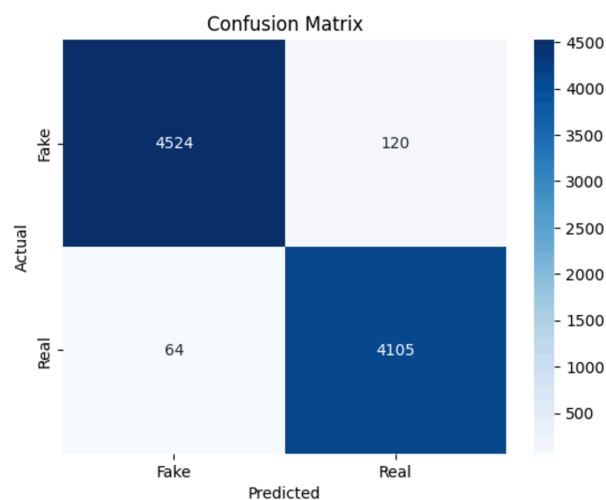
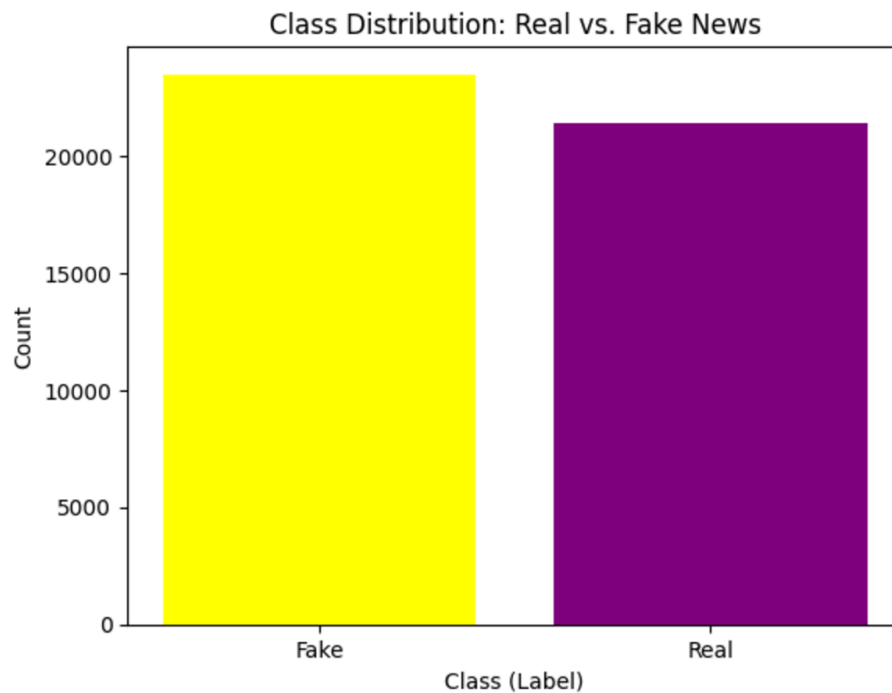
Several machine learning models were trained on the dataset, including Logistic Regression, Decision Trees, Random Forest, and Support Vector Machine (SVM), and their performances evaluated using accuracy, precision, recall, and F1-score metrics. Random Forest was identified as the best performing algorithm amongst these models as it was able to handle the high-dimensionality of the textual features efficiently and was also less prone to overfitting. It attained the best accuracy and F1-score, which embodies descriptive capabilities showing relatively no imbalance between false and normal news articles. The random forest model is performing better, but this is mainly attributed to the nature of a random forest where multiple decision trees work together to better predict an accuracy. On the contrary, simple models like Logistic Regression are based on linear assumptions. Random Forest is known to work well for very complicated non-linear patterns which is the case commonly for textual data. In addition, its ability to rank features according to their importance also allowed for determining which text and metadata attributes contributed the most to the classification task.

Meanwhile, some models such as SVM and Logistic Regression despite being effective, had slightly reduced performance specifically recall. These models, on the other hand, faced challenges with subtle differences and variability in the dataset, especially in the case of borderline cases of real vs fake news. The Random Forest remains the most appropriate model for this problem, having shown

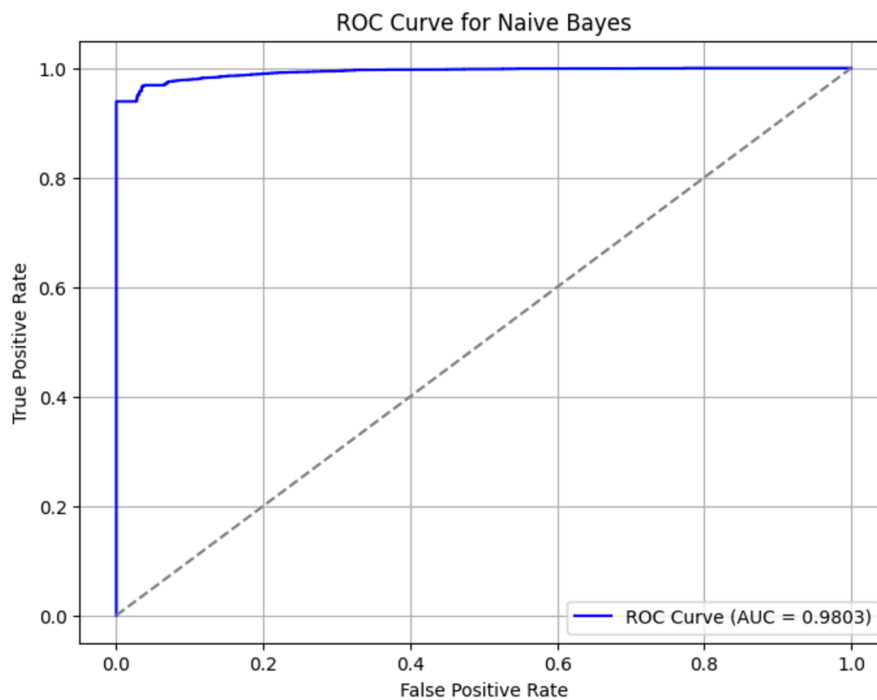
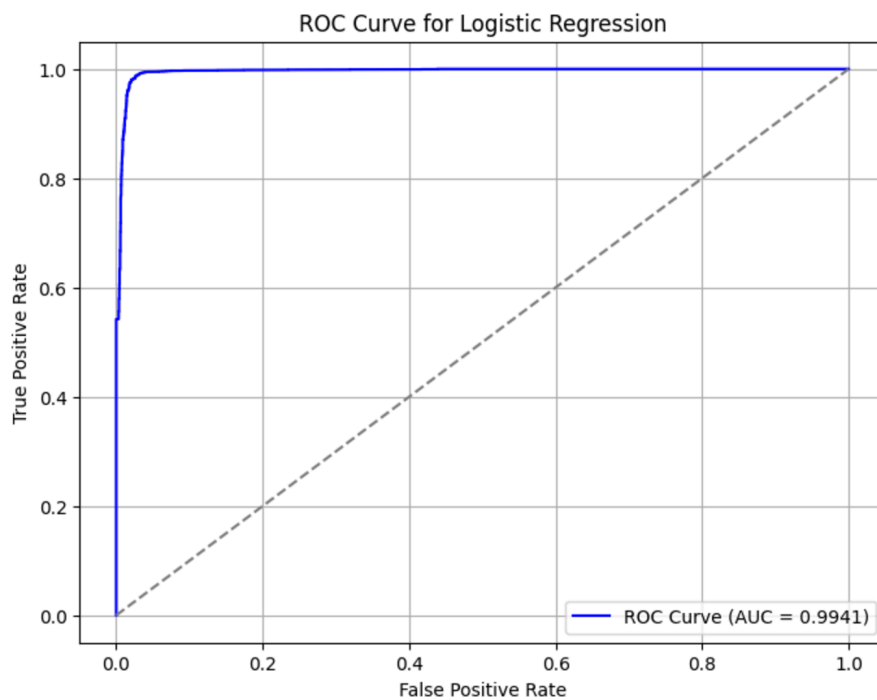
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higher metric scores overall, and a balance of interpretability and computational efficiency, compared to the other models.

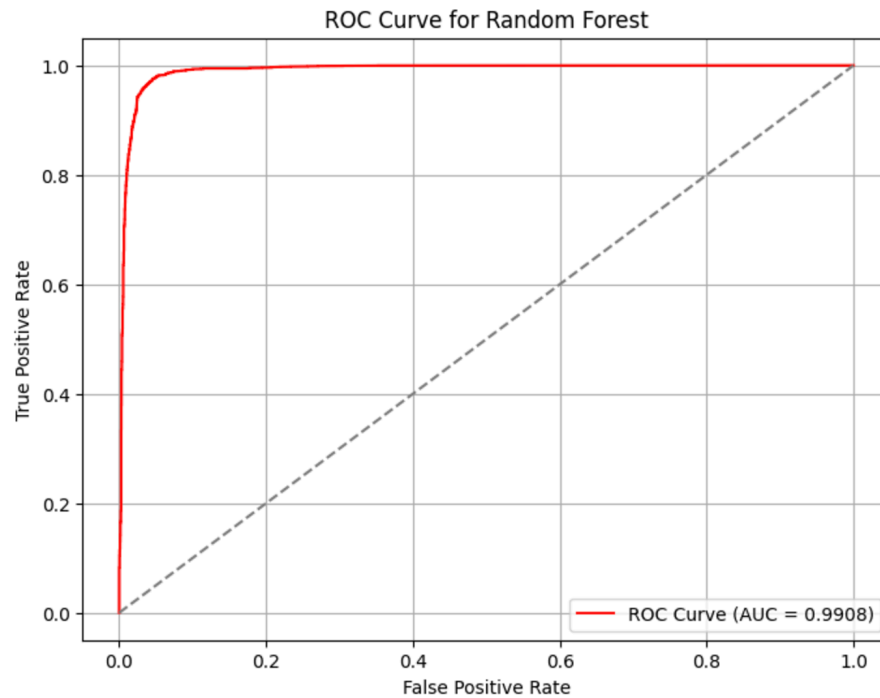
8) Data Visualization.



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9) Conclusion (1 Paragraph).

The Fake News Detection project successfully demonstrates the application of machine learning techniques to address the growing challenge of misinformation. By using PySpark and machine learning algorithms, the project achieved a robust classification system capable of distinguishing real news from fake news with high accuracy. Among the evaluated models, Random Forest emerged as the most effective due to its ensemble nature and ability to handle complex data patterns. This study highlights the importance of proper data preprocessing, feature extraction, and model evaluation in building reliable AI solutions.

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10) Relevant References

<https://colab.research.google.com/drive/1FwcFlbLi5f7WFGfdPytLHskq1N-YQW-A?usp=sharing#scrollTo=BYgQzcDeQ-8L>